

Healthy Vibration Check Completed — Cooling Pump B

Asset: AST_PMP_002 (Pump) | Bearing: BRG_007 (SKF6208)
Fault Mode: healthy | ISO Stage: No Stage (Healthy/Sensor)
Learned Case — Knowledge Agent Retrieval Document | DRO Baseline
CASE_006 | Outcome: SUCCESSFUL | Opened: 2026-05-08 | Closed: 2026-05-08

1. Detection Summary

Detection Date	2026-05-08
Detected By	scheduled_inspection (operator-initiated)
Anomaly Score	0.18 (below alert threshold, well within normal range)
vib_rms_mm_s	2.0 mm/s (within normal range for this pump)
kurtosis	2.0 (well below fault threshold)
temp_c	57°C (within normal range)
bpfo_energy	0.5× (sub-threshold, healthy)
signal_quality_score	1.00
last_lubrication_date	2026-05-05 — 3 days before check, very recent

2. Agent Diagnosis

Primary Diagnosis	healthy — no fault
Confidence	0.94
Reasoning	All signals well within healthy range. No threshold crossed. Lubrication recently completed (3 days prior).
Differential	None required.

3. Action Taken

Recommended Action	Vibration inspection only — no mechanical work required
LOTO Required	No — external vibration inspection, no panel access required
Crew Assigned	K. Mensah (TECH_002) — portable vibration meter walk-around
Work Order	WO_006 — priority LOW

4. Findings at Inspection / Teardown

External walk-around inspection. No mechanical disassembly. Portable vibration readings confirmed the historian data at 2.0 mm/s. Bearing housing was cool to the touch. No visible leaks or abnormal sounds. BRG_008 (NDE) external surface also inspected — no anomalies. Asset confirmed in normal condition. Pump B is in noticeably better condition than Pump A (AST_PMP_001), which had a lubrication issue — Pump B's recent lube service (2026-05-05) has maintained healthy condition.

5. Outcome

Vibration Confirmed	2.0 mm/s (healthy)
Temperature Confirmed	57°C (healthy)

QA Result	PASS
Actual Duration	1.2 hours
Actual Cost	INR 1,800
Recommendation Followed	Yes

6. Root Cause Confirmed

Root Cause	no_fault_found
Notes	AST_PMP_002 healthy baseline confirmed.

7. Lessons Learned

- AST_PMP_002 healthy baseline: approximately 2.0–2.2 mm/s vib and 55–58°C. Any reading above 3.5 mm/s or above 68°C warrants increased monitoring.
- For low-criticality Zone A assets, an external vibration inspection without LOTO is appropriate as a first check. If anomaly confirmed, full work order with LOTO (LOTO_006) is then raised.
- Pump B is in better condition than Pump A — Pump A's lube interval should be monitored more carefully to prevent recurrence of CASE_002.
- Note: LOTO_006 is the correct permit for AST_PMP_002 — separate from LOTO_002 which covers AST_PMP_001.